

1 Solution — GIAM 1.4

Problem 15

1.1 Problem

Calculate the following binomial coefficients: (a) $\binom{3}{0}$; (b) $\binom{7}{7}$; (c) $\binom{13}{5}$; (d) $\binom{13}{8}$; (e) $\binom{52}{7}$.

1.2 The Definition

The binomial coefficient counts the number of k -element subsets of an n -element set:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n(n-1)\cdots(n-k+1)}{k!}.$$

The right-hand “multiplicative” form — k factors on top, $k!$ on the bottom — is the practical way to compute by hand, avoiding gigantic factorials.

1.3 The Five Values

(a) $\binom{3}{0}$. Choosing 0 objects: there is exactly one way (the empty selection). $\binom{3}{0} = 1$.

(b) $\binom{7}{7}$. Choosing all 7 objects: again exactly one way (take everything). $\binom{7}{7} = 1$.

(c) $\binom{13}{5}$. Five factors over $5!$:

$$\binom{13}{5} = \frac{13 \cdot 12 \cdot 11 \cdot 10 \cdot 9}{5!} = \frac{154440}{120} = 1287.$$

(d) $\binom{13}{8}$. By symmetry (below), $\binom{13}{8} = \binom{13}{5} = 1287$.

(e) $\binom{52}{7}$. Seven factors over $7! = 5040$:

$$\binom{52}{7} = \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48 \cdot 47 \cdot 46}{5040} = \frac{674,274,182,400}{5040} = 133,784,560.$$

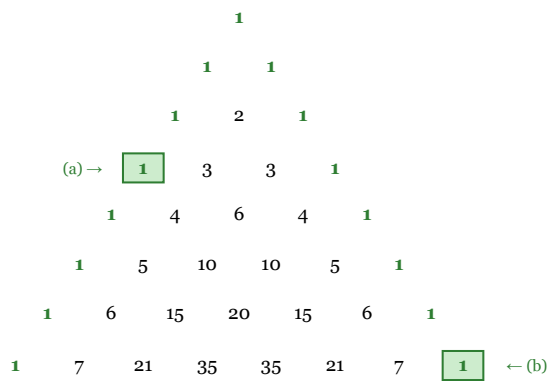
1.4 Summary

Part	Coefficient	Value
(a)	$\binom{3}{0}$	1
(b)	$\binom{7}{7}$	1
(c)	$\binom{13}{5}$	1287
(d)	$\binom{13}{8}$	1287
(e)	$\binom{52}{7}$	133,784,560

Table 1.1

1.5 Reading the Answers Off Pascal’s Triangle

Pascal’s triangle: the edges are all 1 (so $C(n,0) = C(n,n) = 1$), and each row reads the same backwards.



(*c*), (d) use the symmetry $C(n, k) = C(n, n - k)$: since $5 + 8 = 13$,

$$C(13, 5) = C(13, 8) = 1287.$$

(e) $C(52, 7) = 133\,784\,560$.

Figure 1.1: The triangle’s two outer edges are entirely 1’s, so $C(n,0) = C(n,n) = 1$ — covering (a) $C(3,0)$ and (b) $C(7,7)$ at the left and right edges of rows 3 and 7. Each row is left-right symmetric, which gives $C(13,5) = C(13,8) = 1287$, and $C(52,7) = 133\,784\,560$.

1.6 Remark — The Edge Cases $\binom{n}{0} = \binom{n}{n} = 1$

Parts (a) and (b) are the two “edges” of every row of Pascal’s triangle. They are 1 for a simple combinatorial reason: there is exactly **one** way to choose *nothing* from a set ($\binom{n}{0}$ — the empty subset), and exactly **one** way to choose *everything* ($\binom{n}{n}$ — the whole set). Algebraically both fall straight out of the formula, using $0! = 1$:

$$\binom{n}{0} = \frac{n!}{0!n!} = 1, \quad \binom{n}{n} = \frac{n!}{n!0!} = 1.$$

1.7 Remark — The Symmetry $\binom{n}{k} = \binom{n}{n-k}$

Parts (c) and (d) are *equal* — and not by coincidence. The identity

$$\binom{n}{k} = \binom{n}{n-k}$$

holds because choosing which k elements to *include* is the same as choosing which $n - k$ to *exclude* — every k -subset pairs with its complementary $(n - k)$ -subset. Here $\binom{13}{8} = \binom{13}{13-8} = \binom{13}{5}$, so once $\binom{13}{5} = 1287$ is computed, (d) is free. This symmetry is why each row of Pascal’s triangle is a palindrome, and choosing the *smaller* of k and $n - k$ always minimises the work — for (d), computing $\binom{13}{5}$ (five factors) instead of $\binom{13}{8}$ (eight factors).

1.8 Remark — Use the Multiplicative Form, Not Factorials

Computing $\binom{52}{7}$ from $\frac{52!}{7!45!}$ directly would mean handling $52!$, a 68-digit monster. The multiplicative form sidesteps this: only 7 factors on top, cancelled against $7! = 5040$. Even better, cancel *as you go* to keep the running product small — e.g. $\binom{52}{7} = \frac{52}{4} \cdot \frac{51}{3} \cdot \frac{50}{5} \cdot \frac{49}{7} \cdot \frac{48}{2} \cdot 47 \cdot \frac{46}{1}$ (pairing each denominator factor with a numerator it divides). The answer 133,784,560 stays well within reach of pencil-and-paper. This “cancel early” habit is the standard way to evaluate binomial coefficients by hand.

1.9 Remark — $\binom{52}{7}$ Counts Seven-Card Hands

Binomial coefficients are *the* counting tool for unordered selections, and $\binom{52}{7}$ has a concrete meaning: it is the number of distinct 7-card hands dealable from a standard 52-card deck — the relevant count when evaluating seven cards (as in seven-card stud, or the best-five-of-seven in Texas hold'em). So there are **133,784,560** such hands. (Its more famous cousin is $\binom{52}{5} = 2,598,960$, the number of five-card poker hands.) The “order doesn’t matter” feature of $\binom{n}{k}$ is exactly right for card hands, where $\{A\heartsuit, K\spadesuit\}$ is the same hand however it was dealt.

1.10 Remark — Pascal’s Rule Connects Adjacent Rows

A final structural note: every interior entry of Pascal’s triangle is the sum of the two above it,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k},$$

which is how the triangle is *built* row by row (and is provable by the “fix one element: either it’s in the subset or it isn’t” argument). For instance $\binom{13}{5} = \binom{12}{4} + \binom{12}{5} = 495 + 792 = 1287$ — a way to get (c) without any multiplication at all, just by extending the triangle. The edge rule (a, b), the symmetry (c = d), and Pascal’s rule together let you generate every binomial coefficient purely additively.